

Zainab Tairu

✉ tairuzainab5@gmail.com —  LinkedIn —  Github —  Website —  Lagos, Nigeria

Bio

I am a Machine Learning Engineer specializing in NLP and AI accessibility, passionate about optimizing systems and creating solutions for low resource spaces.

Skills

Hard Skills

- Deep Learning** Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs)
- NLP** Named Entity Recognition (NER), Sentiment Analysis, Transformer Models, Rasa, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs).
- Computer Vision** Image Classification
- Languages** Python, SQL, Cypher
- Mathematical Analysis** Linear Algebra, Calculus, Probability and Statistics
- Libraries** Pytorch, spaCy, NLTK, Sckit-Learn, HuggingFace, Tensorflow
- Web Frameworks** FastAPI
- Version Control** Git, GitHub

Soft Skills

Communication, Teamwork, Problem-Solving, Leadership, Adaptability, Creativity, Presentation, Attention to Detail, Time Management.

Experience

The Innoverse (Sandton, South Africa) **Oct 2024 – Present**

NLP intern

- Led the design, development, and launch of the company’s WordPress website, improving user experience and boosting brand visibility.
- Designed an interactive chatbot aimed at streamlining customer service interactions, pending deployment.
- Designed a knowledge graph to improve the response accuracy and user engagement of the chatbot.
- Integrated Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) in the chatbot to significantly improve response quality and personalization.
- Trained Text to speech models for English and Zulu pending deployment.

Hamoye (Lagos, Nigeria) **Nov 2023 – Dec 2023**

Intern

- Worked under the guidance of industry experts to train a predictive model to assess the impacts of climate change on Agricultural products
- Wrote a research proposal for the Prediction of African conflicts advocating for proactive strategies based on our analytical insights to guide crucial policy decisions.
- Designed a PowerPoint presentation highlighting the potential risks of food-related conflicts in African nations to a panel of judges.

Publications (In Progress)

Multi-attention mechanisms with Transformer based NER models for Resume Parsing **Jan 2024 – Present**

Co-authored with Dr. Avashna Govender

- Co-authored a research paper on improving Named Entity Recognition (NER) for resume parsing using multi-attention mechanisms with transformer-based models.
- Developed an innovative multi-attention mechanism that significantly enhanced the model’s ability to identify and contextualize key resume elements such as skills, certifications, and educational background.

Sequence Analysis and Engineering of Bacterial Lipases for Industrial Applications **Jan 2025 – Present**

- Developing a custom ML framework to predict and generate entirely new bacterial lipases from sequence data, with a focus on applications in Nigerian industries.
- Combining sequence clustering, evolutionary analysis, and computational protein engineering to create lipases with enhanced specificity and stability, with a potential for local scalability and cost-effective production.

Self Development Projects

LET ME COOK

Nov 2024

- Originated and refined a unique dataset of a diverse collection of Nigerian and globally inspired recipes adapted to appeal to the Nigerian palate from leading culinary websites and personal expertise to ensure authenticity and relevance.
- Deployed on streamlit with a specific aim to empower students to efficiently utilize their pantry items here

ChemSum

Nov 2024 – Present

- Developing an NLP-based tool to automatically extract and summarize key sections (Abstract, Methodology, Key findings, Results, limitations, safety considerations) from Chemical Engineering research papers.
- Manually Annotated 50 Chemical Engineering Research papers
- Integrating audio functionality using Google's Text-to-Speech API to enhance accessibility.

Education

Bachelor of Science in Chemical Engineering

Obafemi Awolowo University, Osun, Nigeria

Jan 2022 – Present

Expected Graduation: August 2026

Volunteering Work

Star Girls Africa

Jun 2024 – October 2024

Data science tutor

- Authored a comprehensive beginner's curriculum in data science.
- Delivered training to 100 students, enhancing their foundational skills.
- Achieved notable increase in student engagement and skills proficiency

Hackathon

SPE Hackathon(Osun State, Nigeria)

January 2025

- Developed a machine learning model using a Self-Attention Convolutional Neural Network (SACNN) to enhance porosity prediction in reservoir characterization.
- Integrated well log and seismic data for accurate model training, achieving a Mean Squared Error (MSE) of 0.00337 and a Root Mean Squared Error (RMSE) of 0.05713.
- Authored a detailed paper on the methodology, challenges, and results of implementing machine learning for geological assessments in Nigeria's oil sector

Conferences/Workshops

Deep Learning Indaba(Dakar, Senegal)

September 2024

- Participated in workshops and sessions on AI ethics, transformer models, and low-resource NLP, gaining hands-on experience with cutting-edge machine learning techniques.

Data Science Network AI Bootcamp(Lagos, Nigeria)

October 2024

- Participated in a hackathon to develop an English-to-Yoruba translation model, addressing NLP challenges for low-resource languages.

References

Dr. Avashna Govender

Research Group Leader, CSIR

✉ avashna002@gmail.com

Mr. Simphiwe Zitha

Deep Learning Indaba Mentor

✉ nhlanhla34@gmail.com